Engr. Jhaycee Anthony P. Acain

Master of Science in Computer Engineering Major in Digital Hardware Systems

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September 16, 2001



EDUCATIONAL BACKGROUND

2024 – Present

Master of Science in Computer Engineering - Major in Digital Hardware Systems
University of San Carlos - Talamban Campus
Nasipit, Talamban, Cebu City, 6000

2020 – 2024

Bachelor of Science in Computer Engineering
University of San Carlos - Talamban Camopus
***GPA*: 1.46**
Academic Awards*: *Cum Laude
Nasipit, Talamban, Cebu, City, 6000

NOTABLE ACADEMIC ACHIEVEMENTS

University Scholar

- Department of Science and Technology (DOST) - Engineering Research and Development for Technology (ERDT) Scholarship Program (2024 - Present)
- Yap Yok, Jr. and Betty Yap School of Engineering Charity Foundation Scholarship (2022 - 2024)
- Liloan Scholarship Program (2018 - 2024)

1st Runner-Up

- 7th Lear Innovation Olympics 2023 held at Lear Corporation, MEPZ II, Basak, Lapu Lapu City, 6015, Cebu

PROFESSIONAL EXPERIENCE

08/2024 – Present

Master's Thesis Researcher
University of San Carlos

- **Engineered** low-level C firmware and embedded hardware interfaces, optimizing DMA register configurations and peripheral communications to achieve high-speed data acquisition for an Analog to Digital Converter (ADC) functionality.
- **Developed** a real-time data visualization pipeline using Go and WebGL, streaming high-frequency signals via WebSockets for low-latency signal rendering for developing a browser-based Oscilloscope.
- **Validated** system performance across test cases, securing proposal approval and presenting research findings at academic conferences

2025 – Present

Member
Japane Society of Mechanical Engineers (*JSME*)

- **Participated** in Robotics and Mechatronics (RoboMech) 2025 Conference in Yamagata Big Wing, Yamagata City, Yamagata Prefecture, Japan
- **Enhanced** knowledge exchange through contributing insightful suggestions and discussions to poster presenters.

04/2026

Academic Research Conference Poster Presenter
10*th University of San Carlos* (*USC*) *University Research Conference* (*URC* 2026)

- **Presented** research findings on "*Performance Assessment and Proposed Reimplementation of Lab in A Box Software*" to university faculty, panel members, and academic peers.

03/2026

Academic Research Conference Oral Presenter
Robotics and Automation Conference / Computer Engineering and Technology Conference (*RACCETCON*) 2026

- **Delivered** an oral presentation on "*Lab in A Box*: *Performance Assessment and Reimplementation of LABSoft*" at the Robotics and Automation Conference / Computer Engineering and Technology Conference.

01/2025 – 12/2025

Assistant Research Adviser
University of San Carlos

- **Co-Advised Research & Hardware/Software Co-Design:** Assisted in advising and mentoring undergraduate computer engineering students on the design, architecture, and development of the *Lab in a Box* (*LAB v3*) portable electronics trainer kit.

- **System Integration & Signal Processing:** Guided student development of an Automated Analog Circuit Checker within LABSoft utilizing C++ and FLTK, integrating cross-correlation and Mean Squared Error (MSE) algorithms to perform real-time waveform similarity analysis against reference templates.
- **Embedded Firmware & Hardware Testing:** Supervised the implementation of an SPI-based multi-microcontroller architecture offloading input acquisition and logic analyzer sampling to a Raspberry Pi Pico (RP2040) to interface with a Raspberry Pi Zero W master controller.
- **Performance Assessment & Experimental Validation:** Overseen hardware verification, theoretical latency benchmarking (achieving ~69.67ms UI navigation response times), and comparative signal fidelity testing against commercial test instruments (Analog Discovery 2).

08/2025 **Academic Research Conference Poster Presenter**
National Electrical, Electronics, and Computer Engineering Conference (*NEECECON*) 2025
- **Communicated** technical methodology and research outcomes for "*Lab in A Box v*3: *Development of Analog Circuit Checker, Software Navigation Module and Reimplementation of LABSoft*" at the National Electrical, Electronics, and Computer Engineering Conference.
- **Articulated** complex and technical descriptions and findings clearly and effectively.
- **Provided** concise summaries of my research to interested parties.

10/2024 – 05/2025 **Academic Research Assistant**
Emerson Electric Co.
- **Modeled** a Perceptron Artificial Neural Network (ANN) for machine status assessment and decide execution based on predefined activation threshold using Python Programming Language.
- **Trained** the Perceptron model on 625 unique raw data points for accurate performance prediction.
- **Trained** the machine learning model to conduct informed decision-making based on pattern recognition.

01/2024 – 05/2024 **Software Automation Engineer Intern**
Techflow.ai (*German-based Automation Company*)
- **Developed** an automated invoice creation system, streamlining financial operations and reducing manual data entry.
- **Created** an automation process for real-time sales data updates and processing, improving reporting accuracy.
- **Improved** data validation and processing with Regular Expressions (RegEx) in an automation design.

2022 – Present **Computer Engineering Laboratory Staff**
Printed Circuit Board (*PCB*) *and Center for Robotics and Automation* (*CRA*) *Laboratory at University of San Carlos - Talamban Campus*
- **Designed** and **prototyped** PCBs, ensuring all specifications were met for fabrication.
- **Manufactured** and **assembled** PCBs, managing the end-to-end process from design to a final, functional board using KiCad.
- **Modeled** and **printed** 3D parts for robotics projects, creating custom components to meet specific project requirements using Solidworks and Bambu Studio Software Application.

2022 – Present **Freelance Software Developer**
- **Developed** an application for an automated laundry service, featuring a weight-based cost calculator and user notifications upon completion of wash and dry cycles using C++ for the hardware, Node.js for the backend, Javascript for the frontend, and MySQL for the database.
- **Implemented** a smart laundry solution by integrating a weight sensor to calculate costs and a notification system to alert users when a wash and dry cycle is complete.
- **Developed** a custom Point of Sale (POS) application for a water refilling station, which streamlined their sales tracking, inventory, and customer management using MongoDB as databased and Java as frontend and backend.

08/2023 **Academic Research Booth Exhibitor**
Technology Advocacy (*Techvocacy*) *Expo* 2023 *organized by Ramon Aboitiz Foundation Inc.* (*RAFI*)
- **Managed** the University of San Carlos research booth at the event, engaging with over 200 attendees to promote our work in Computer Engineering.
- **Showcased** a prototype, software, or system to booth visitors.
- **Highlighted** the capabilities, achievements, and goals of the different research in the institution.
- **Established** connections with peers, industry professionals, and academic leaders.